

Java Programming Assignment

Focus: Object-Oriented Programming & Design Patterns

Project: Banking System Core

Develop a robust backend engine for a Banking Application emphasizing OOP principles.

1. Object-Oriented Design (Classes & Objects)

- Create an abstract class `Account` with attributes: `accountNumber`, `balance`, and `owner`.
- Implement subclasses: `SavingsAccount` (with interest rates) and `CheckingAccount` (with overdraft limits).
- Use **Encapsulation** by making fields private and providing getters/setters.

2. Interfaces & Polymorphism

- Define an interface `Transaction` with a method `execute()`.
- Implement `Deposit` and `Withdrawal` classes that implement the `Transaction` interface.
- Ensure the system can process a list of different transaction types polymorphically.

3. Exception Handling & Logic

- Create a custom exception `InsufficientFundsException`.
- Throw this exception if a withdrawal exceeds the available balance or overdraft limit.
- Use `try-catch-finally` blocks to handle transaction logging.

4. Collections Framework

- Use a `HashMap` to store and retrieve accounts by their ID.
- Use `Streams API` to filter accounts with a balance greater than \$10,000.

5. Bonus: Multithreading

- Simulate concurrent deposits to the same account using `synchronized` methods or `ReentrantLock` to prevent data inconsistency.